Project Design Phase-I
Proposed Solution

Date	09 March 2026
Team ID	698c67ecb31ff1dfca225748
Project Name	PlantCare AI: Advanced Plant Disease Detection Using Transfer Learning
Team Members	Akshat Bhati, Shubhadipta sarkar, Haridharshan B, Madan M
Maximum Marks	2 Marks

Proposed Solution

Project team has filled the following parameters based on the PlantCare AI project.

S.No	Parameter	Description
1	**Problem Statement (Problem to be solved)**	Agricultural plant diseases destroy 20–40% of global crop production annually, costing the world economy over $220 billion USD in losses. The core problem has three dimensions: 1. EXPERTISE GAP: Accurate plant disease diagnosis requires trained agronomists — professionals who are expensive, geographically constrained, and unavailable in real-time for most of the world's 500 million smallholder farming households. 2. TIME CRITICALITY: Plant diseases spread exponentially. A 24–72 hour delay in diagnosis leads to outbreak-level crop loss. Manual lab-based testing takes days. 3. ACCESSIBILITY: Farmers in developing regions (Sub-Saharan Africa, South Asia, Latin America) have limited access to agricultural extension services. A solution that requires specialized hardware or expert users fails these communities.
2	**Idea / Solution Description**	PlantCare AI is an end-to-end AI-powered plant disease detection web application built using Transfer Learning on the MobileNetV2 convolutional neural network architecture. **Phase 1 — Model Training (Google Colab):** The New Plant Diseases Dataset (87,000+ images, 38 classes) is downloaded from Kaggle. MobileNetV2 pre-trained on ImageNet-1000 is loaded without its classification head. The last 10 layers are unfrozen for domain-specific fine-tuning. A custom classification head is added: GlobalAvgPool2D → Dropout(0.35) → Dense(256, ReLU) → Dropout(0.25) → Dense(38, Softmax). The model is compiled with Adam (lr=1e-4) and Categorical Crossentropy, trained for 5 epochs with

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		ModelCheckpoint, ReduceLROnPlateau, and EarlyStopping callbacks. **Phase 2 — Web Application (Flask / Local):** The trained model (.keras) and class names (.json) are integrated into a Flask application with 5 routes and 4 HTML pages. Users upload a plant leaf photograph — the system preprocesses it (resize to 224×224, normalize), runs inference through the model, and displays the disease classification with health status badge and recommended actions.
3	**Novelty / Uniqueness**	• Deployment-Optimized Architecture: MobileNetV2 was deliberately chosen over heavier models (VGG16 138M params vs MobileNetV2 3.4M params). This 40× size reduction enables sub-1.5-second CPU inference, making the solution deployable on any standard server without GPU infrastructure — dramatically reducing hosting cost compared to competing solutions. • Zero-Configuration Deployment: Unlike research-oriented plant disease models that require GPU hardware, Python expertise, or cloud API subscriptions, PlantCare AI runs on a standard laptop with pip install. Any user with Python can deploy it in under 5 minutes. • Breadth of Coverage: 38 classes across 14 plant species — including Apple, Tomato, Corn, Grape, Potato, Pepper, Peach, Cherry, Strawberry, Blueberry, Orange, Raspberry, Soybean, and Squash — making it one of the most comprehensive open-source plant disease classifiers available as a ready-to-deploy web application. • Full-Stack Integration: Unlike standalone Kaggle notebooks, PlantCare AI delivers the complete ML-to-UI pipeline — from raw dataset to a live, interactive web application usable by non-technical end users.
4	**Social Impact / Customer Satisfaction**	• Smallholder Farmers (Primary Beneficiaries): 500M+ smallholder farming households globally gain access to expert-level plant pathology diagnosis at zero cost. Early detection enables targeted, low-cost treatment rather than entire crop abandonment. • Food Security (UN SDG 2 — Zero Hunger): Reducing crop disease losses directly supports global food security. Even a 10% reduction in disease-related crop loss translates to billions in saved food production value. • Reduced Chemical Overuse: Accurate diagnosis prevents misidentification — farmers over-apply broad-spectrum pesticides when uncertain. Precise diagnosis enables targeted treatment, reducing chemical runoff and environmental damage (aligns with UN SDG 15 — Life on Land). • Agricultural Education: The full training notebook serves as a hands-on AI/ML learning resource for agricultural and computer science students. Customer satisfaction is high — zero technical knowledge required to use the web interface.

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5	**Business Model (Revenue Model)**	**Freemium SaaS Model:** • Free Tier: Unlimited basic disease diagnosis for individual farmers and home gardeners — funded by the project as a social impact initiative. • Enterprise API Subscription: Monthly subscription for agri-businesses, food processors, and drone-monitoring companies requiring high-volume API access (1000+ predictions/day), batch processing, and SLA guarantees. Pricing: $99–$999/month by tier. • White-Label Licensing: License the trained model and codebase to agri-tech companies building their own branded plant health products. One-time licensing fee + annual maintenance. • Government / NGO Partnerships: Subsidized deployment partnerships with agricultural ministries in developing nations (World Bank, FAO funded programs). • Premium Mobile App: Future paid mobile application with offline TensorFlow Lite inference for zero-connectivity remote farm use. Subscription: $2.99/month.
6	**Scalability of the Solution**	• Horizontal Scaling: Flask + Gunicorn is Docker-containerizable. Kubernetes orchestration enables auto-scaling based on real-time traffic — from 10 to 10,000 concurrent users without code changes. • Model Scalability: MobileNetV2 model can be converted to TensorFlow Lite for mobile deployment, or served via TensorFlow Serving / AWS SageMaker for dedicated GPU-backed inference at enterprise scale (1000+ requests/second). • Dataset Expansion: Training pipeline designed to retrain on expanded datasets — adding new plant species or disease classes requires only adding training images and re-running the Colab notebook. No architectural changes needed. • API-First Architecture: Flask routes can be extended to expose a REST API (JSON responses) enabling integration with mobile apps, IoT sensors, drone systems, and third-party agricultural platforms without UI changes. • Global CDN Delivery: Static assets (CSS, JS, images) served via CDN in production — zero latency for users worldwide regardless of server location.